

Critical Metrology

Parameter Estimation limits near Quantum Phase Transitions in the Quantum Rabi Model

Lander Zulueta Loizaga

Faculty of Science and Technology
University of the Basque Country (UPV/EHU)

Advisors: Iñigo L. Egusquiza & Jagoba Apellaniz

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- ① Introduction to Critical Metrology
- ② Analytical Approach
- ③ Numerical Methods across the Phase Transition
- ④ Conclusions & Future Outlook

What is Metrology?

Metrology is the science of measurement. We focus on **parameter estimation** in the *asymptotic limit* (large sample size case).

Classical Parameter Estimation

Precision bounded by the **Shot-noise limit**:

$$(\Delta g)^2 \sim \frac{1}{\nu N}$$

where N is the size of the sample and ν the number of repetitions.

Quantum Parameter Estimation

Exploiting quantum resources allows reaching the **Heisenberg limit**:

$$(\Delta g)^2 \sim \frac{1}{\nu N^2}$$

A quadratic improvement over the classical case.

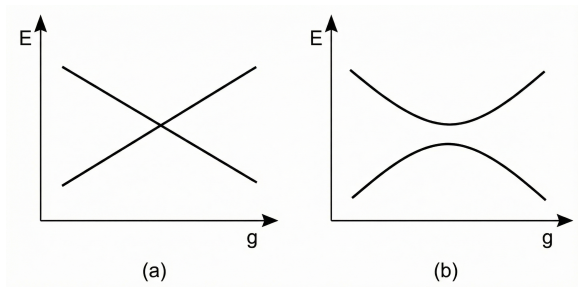
Critical Metrology & Quantum Phase Transitions

Critical Metrology:

- Exploits **extreme susceptibility** near a phase transition for enhanced parameter estimation.

Quantum Phase Transition (QPT):

- Occurs at $T = 0$.
- Driven by **quantum fluctuations**.
- Signalled by the **closing of the energy gap**.
- Strictly requires the **thermodynamic limit** ($N \rightarrow \infty$).



Energy gap closing at a QPT vs. avoided level-crossing.

Precision Metric: Quantum Fisher Information

How do we quantify estimation precision?

Quantum Cramér–Rao Bound

The variance of *any* unbiased estimator of g is bounded by:

$$(\Delta g)^2 \geq \frac{1}{\nu F_Q(\varrho_g)}$$

where ν is the number of independent measurements and F_Q is the **Quantum Fisher Information** (QFI).

- **Our goal:** Show that F_Q diverges at the critical coupling g_c (w/o considering critical slowing down)
- For **pure states**:

$$F_Q = 4 \left[\langle \partial_g \psi | \partial_g \psi \rangle - |\langle \partial_g \psi | \psi \rangle|^2 \right]$$

The Quantum Rabi Model (QRM)

A single two-level atom coupled to one cavity mode:

$$H_{\text{QRM}} = \omega a^\dagger a + \Delta \sigma_z + g \sigma_x (a + a^\dagger)$$

- ω : cavity frequency, 2Δ : qubit splitting, g : coupling strength.

Classical Oscillator Limit

The QRM achieves critical behavior with a *single* atom when

$$\frac{\Delta}{\omega} \longrightarrow \infty.$$

The frequency ratio plays the role of a *thermodynamic limit*. Exhibits a *Superradiant* QPT with order parameter $\langle a^\dagger a \rangle \frac{\omega}{\Delta}$.

Analytical Approach: Schrieffer–Wolff Transformation

We decouple spin and bosonic subspaces via a unitary $U = e^S$ (S anti-hermitian).

First-order effective projected Hamiltonian:

$$H_{\text{SW1}} = \omega a^\dagger a - \Delta - \frac{g^2}{2\Delta} (a + a^\dagger)^2$$

- Applied a mean-field to find the **critical point**: $g_c = \sqrt{\omega \Delta/2}$
- Recognised GS as a *squeezed vacuum* via a Bogoliubov transformation.

Resulting QFI (Normal Phase):

$$F_Q(\varrho_g) \approx \frac{2g^2}{(\omega \Delta - 2g^2)^2} \xrightarrow{g \rightarrow g_c} \text{diverges}$$

- The **second-order** Schrieffer–Wolff expansion yields the same divergence.

Overview & Limitations of the Analytic Approach

Achievements:

$$F_Q(\varrho_g) \approx \frac{2g^2}{(\omega\Delta - 2g^2)^2} \propto (g_c - g)^{-2}$$

- Derived the QFI divergence *analytically*.
- Found a critical exponent $\nu = -2$.

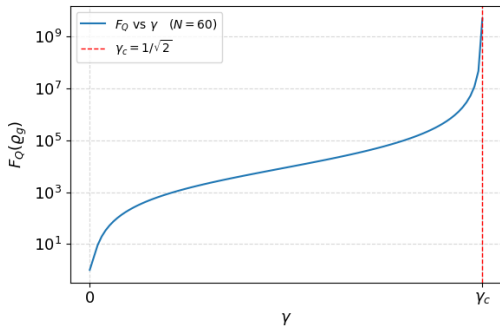
The Superradiant Phase Problem

Beyond g_c the effective potential inverts, a Bogoliubov is not suitable, *numerical* strategy is required. We will work on that after a quick **cross-check**.

Numerical Validation (Normal Phase)

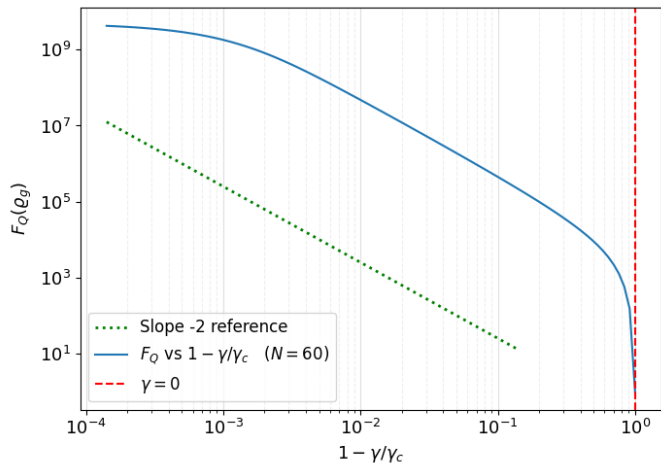
Before crossing the critical point we validate the numerics against the analytical prediction.

- Exact diagonalisation using QuTiP (truncation $N=60$).
- **Result:** Numerical QFI (blue) matches the analytical divergence.
- **Note:** Instead of g , the dimensionless coupling strength $\gamma = \frac{g}{\sqrt{\Delta\omega}}$ will be used onwards. $\gamma_c = \frac{1}{\sqrt{2}}$.



QFI vs. γ ($\omega/\Delta = 5 \times 10^{-5}$, $N=60$).

Critical Exponent in the Normal Phase



Log-log scaling: critical exponent $\nu = -2$ in the Normal Phase.

The Superradiant Collapse

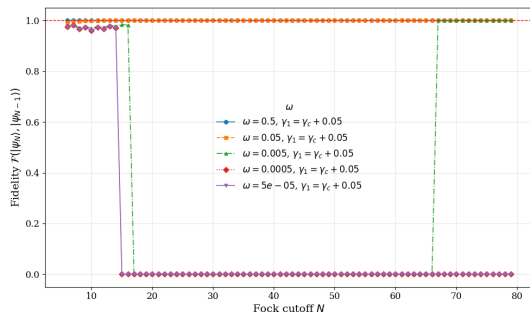
Problem

Crossing γ_c , the standard Fock-basis computation breaks down.

Why does it fail?

- 1 The photon number $\langle a^\dagger a \rangle$ becomes *macroscopic*.
- 2 A Fock basis centred at the vacuum $|0\rangle$ requires $N \rightarrow \infty$ to capture the state.

Consequence: Fidelity with the true ground state drops to zero.



Fidelity collapse in the standard basis beyond γ_c .

Solution: Displaced Squeezed Basis

To capture the superradiant ground state, we *adapt* the Fock basis in two steps.

- 1 **Displacement** — shift the origin to the mean-field amplitude α_s (variational):

$$D(\alpha_s) = \exp(\alpha_s a^\dagger - \alpha_s^* a)$$

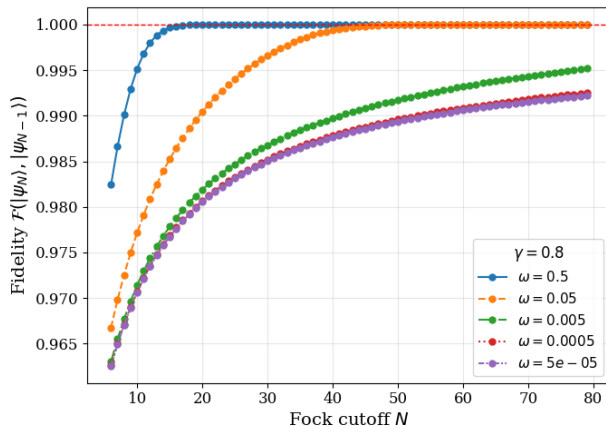
- 2 **Squeezing (Bogoliubov rotation)** — define new bosonic operators $(b, b^\dagger)$ adapted to the potential's curvature.

Result

The new $(b, b^\dagger)$ basis efficiently hosts the macroscopic photon occupation of the superradiant phase.

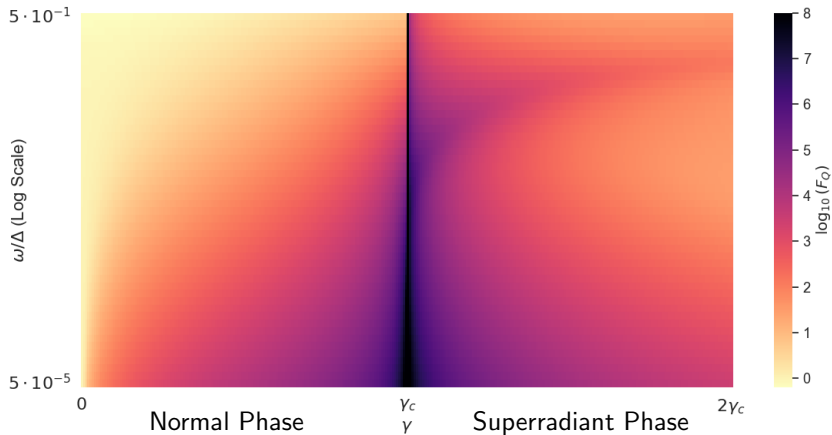
Convergence Restored

Using the displaced squeezed ($b, b^\dagger$) basis, high fidelity is recovered with modest truncation ($N \approx 60$).

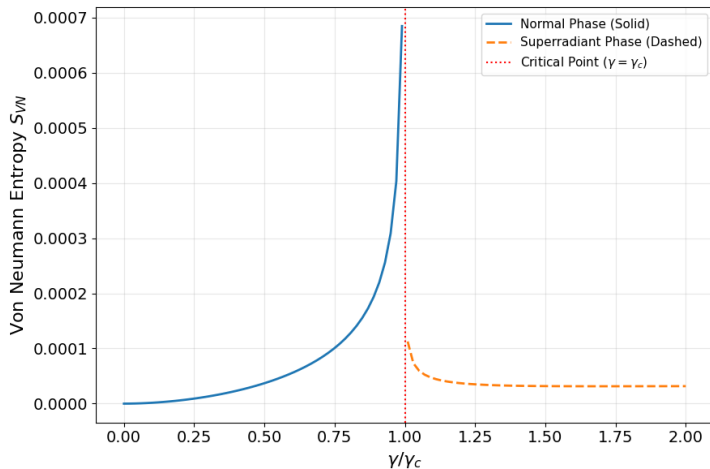


QFI Landscape across the Phase Diagram

Using the displaced squeezed basis, we can map the QFI in *both* phases.



Entanglement Entropy



Entanglement entropy ($\omega/\Delta = 5 \cdot 10^{-5}$): tendency change at γ_c . Validity of mean-field.

Overview & Conclusions

- 1 **Finite-size criticality.** The QRM exhibits thermodynamic-like criticality in the $\Delta/\omega \rightarrow \infty$ limit with a *single* atom.
- 2 **Divergent precision.** The QFI diverges at g_c , confirming the viability of critical metrology.
- 3 **Methodology.**
 - Analytical (Schrieffer–Wolff) methods yield analytical results in the Normal Phase.
 - Displacement + Squeezing enable stable numerics in the Superradiant Phase.

Future directions:

- Study *critical slowing down* (dynamical protocols).
- Confirm intuition behind the anomalous branching of QFI in the superradiant phase.

Thank you!

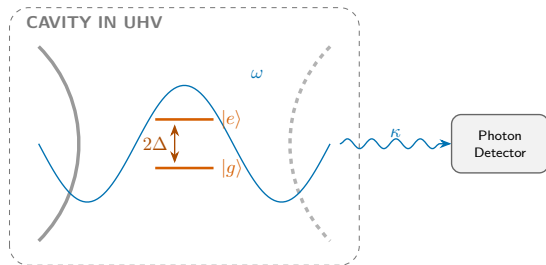


<https://github.com/Landerzl/Critical-Metrology>

Questions & Discussion

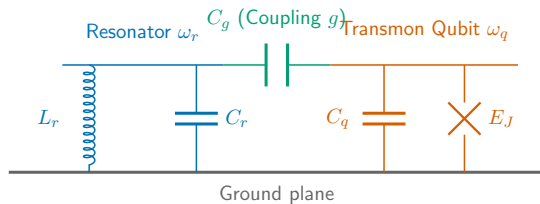
Experimental Implementations of the QRM

Cavity QED (Optical Regime)



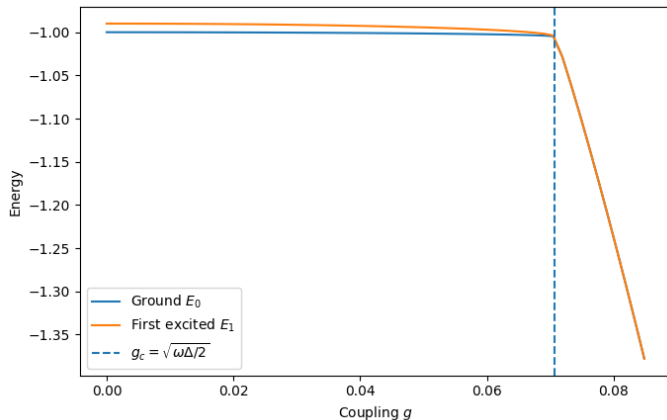
- Natural atoms (e.g., Alkali metals)
- Fabry-Pérot optical resonators
- Typically operates in the weak/strong coupling regimes

Circuit QED (Microwave Regime)



- Artificial atoms (Superconducting circuits)
- Coplanar waveguide (CPW) resonators
- Access to ultrastrong/deep strong coupling

Backup: Spectral Gap Closure



Ground state (E_0) and first excited state (E_1) energies versus coupling. The spectral gap closes exactly at the analytical prediction $g_c = \sqrt{\omega\Delta/2}$.

Inspirational Quote

